

Inorganic Chemistry Practical: Systematic Qualitative Analysis – Cations, Anions & Gases

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5\text{ }^{\circ}\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Systematic Differentiation of Amphoteric Metal Ions: Al^{3+} , Zn^{2+} , and Mg^{2+} **Cambridge Syllabus Reference: 9701/31/M/J/24/Q3 · [10 Marks]**

Three solutions, FA 1, FA 2, and FA 3, contain Al^{3+} , Zn^{2+} , and Mg^{2+} .

Test 1: Add $\text{NaOH}(\text{aq})$ dropwise, then in excess.

Test 2: Add $\text{NH}_3(\text{aq})$ dropwise, then in excess.

(a) Complete the observation table below. [4]

Test	Observations	Deductions
Al^{3+} : (i) NaOH dropwise & excess (ii) NH_3 dropwise & excess		
Zn^{2+} : (i) NaOH dropwise & excess (ii) NH_3 dropwise & excess		
Mg^{2+} : (i) NaOH dropwise & excess (ii) NH_3 dropwise & excess		

(b) Write the formula of the complex ion formed when zinc hydroxide dissolves in excess ammonia. [2]

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(c) Write the balanced ionic equation for the reaction of aluminium hydroxide with excess hydroxide ions. [2]

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(d) Explain why magnesium hydroxide does not dissolve in excess sodium hydroxide. [2]

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Question 2: Precipitation and Complex Ion Ligand Substitution of Copper(II) Ions**Cambridge Syllabus Reference: 9701/35/M/J/24/Q3 · [10 Marks]**

Copper(II) sulfate solution is subjected to tests with aqueous sodium hydroxide, aqueous ammonia, and potassium iodide.

(a) State the observations when aqueous NaOH is added dropwise and then in excess to $\text{Cu}^{2+}(\text{aq})$. [2]

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(b) State the observations when aqueous NH_3 is added dropwise and then in excess to $\text{Cu}^{2+}(\text{aq})$. [3]

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(c) State the formula and geometry of the deep royal blue complex ion. [2]

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(d) State the observations when aqueous potassium iodide, KI, is added to aqueous copper(II) sulfate. [3]

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Question 3: Redox Precipitation and Aerial Oxidation of Iron(II) Hydroxide**Cambridge Syllabus Reference: 9701/32/O/N/23/Q3 · [10 Marks]**

A fresh green solution of iron(II) sulfate is treated with aqueous sodium hydroxide. The test-tube is left open to the atmosphere for 10 minutes.

(a) State the initial observation upon adding aqueous NaOH to $\text{Fe}^{2+}(\text{aq})$. [2]

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(b) Describe the change in appearance of the precipitate after standing exposed to air. [2]

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(c) Explain this color change chemically and identify the species responsible. [3]

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(d) Write the balanced chemical equation for this aerial oxidation. [3]

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Question 4: Alkaline Oxidation of Chromium(III) Ions by Hydrogen Peroxide**Cambridge Syllabus Reference: 9701/34/M/J/23/Q3 · [10 Marks]**

A grey-green solution containing Cr^{3+} is treated with excess aqueous NaOH to form a dark green solution. Aqueous hydrogen peroxide, H_2O_2 , is added and the mixture is warmed gently.

(a) State the formula of the chromium species in the dark green solution before adding H_2O_2 . [2]

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(b) State the observation when the mixture is warmed with hydrogen peroxide. [2]

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(c) Identify the yellow species formed and state the oxidation state of chromium in it. [3]

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(d) Write the balanced redox equation for the oxidation of $[\text{Cr}(\text{OH})_4]^-$ by H_2O_2 in alkaline conditions. [3]

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Question 5: Chemical Discrimination of Sulfate (SO_4^{2-}) and Sulfite (SO_3^{2-}) Ions**Cambridge Syllabus Reference: 9701/31/O/N/23/Q3 · [10 Marks]**

Solution A contains sodium sulfate, Na_2SO_4 . Solution B contains sodium sulfite, Na_2SO_3 .
Both solutions are tested with aqueous barium chloride, BaCl_2 , followed by dilute hydrochloric acid.

(a) State the observations when BaCl_2 is added to Solution A, followed by dilute HCl. [3]

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(b) State the observations when BaCl_2 is added to Solution B, followed by dilute HCl and warming. [4]

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(c) Identify the gas evolved in (b) and write an ionic equation for its reaction with dichromate. [3]

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Question 6: Reduction of Nitrate Ions by Aluminium in Alkaline Solution**Cambridge Syllabus Reference: 9701/33/M/J/22/Q3 · [10 Marks]**

A sample of potassium nitrate, KNO_3 , is dissolved in water. Aqueous sodium hydroxide is added, followed by small pieces of aluminium foil. The mixture is heated gently.

(a) State the observation during heating and describe how the gas evolved is identified. [3]

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(b) State the changes in oxidation number of nitrogen and aluminium in this reaction. [3]

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(c) Write the balanced ionic equation for this reduction. [2]

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(d) Explain why red litmus paper used to test the gas must be damp. [2]

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Question 7: Analytical Verification of Carbonate Ions and Limewater Precipitation**Cambridge Syllabus Reference: 9701/32/M/J/22/Q3 · [10 Marks]**

Solid sodium carbonate, Na_2CO_3 , is treated with dilute hydrochloric acid. The evolved gas is bubbled through limewater, $\text{Ca}(\text{OH})_2(\text{aq})$.

(a) State two observations in the reaction flask when acid is added to the carbonate. [2]

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(b) State the initial observation in the limewater and explain what happens if gas continues to bubble for several minutes. [4]

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(c) Write the ionic equation for the redissolution of calcium carbonate in excess CO_2 and water. [2]

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(d) Describe how to distinguish solid Na_2CO_3 from solid NaHCO_3 using aqueous magnesium sulfate. [2]

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Question 8: Comprehensive Laboratory Protocol for Analytical Gas Detection**Cambridge Syllabus Reference: 9701/34/O/N/22/Q3 · [10 Marks]**

Cambridge 9701 Paper 3 tests six standard laboratory gases: NH_3 , CO_2 , Cl_2 , H_2 , O_2 , SO_2 . Candidates must apply appropriate confirmatory tests and observations.

(a) State the test and positive result for hydrogen gas, H_2 . [2]

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(b) State the test and positive result for oxygen gas, O_2 . [2]

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(c) State the test and positive result for chlorine gas, Cl_2 . [2]

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(d) State the test and positive result for sulfur dioxide gas, SO_2 . [2]

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(e) State the test and positive result for ammonia gas, NH_3 . [2]

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Question 9: Systematic Confirmation of the Ammonium Ion (NH_4^+) in Unknown Salts**Cambridge Syllabus Reference: 9701/35/O/N/23/Q3 · [10 Marks]**

A solid salt contains ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$. A student tests for the cation by warming with aqueous sodium hydroxide.

(a) Describe the step-by-step method and observations to confirm NH_4^+ . [4]

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(b) Write the ionic equation for the reaction producing ammonia. [2]

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(c) Explain why red litmus must NOT touch the rim or inner wall of the test-tube. [2]

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(d) Explain why aqueous ammonia cannot be used as the reagent to test for ammonium ions. [2]

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Question 10: Deductive Identification of an Unknown Binary Inorganic Salt (Salt X)**Cambridge Syllabus Reference: 9701/33/M/J/24/Q3 · [10 Marks]**

An unknown pale green solid salt, X, dissolves in distilled water to form a pale green solution.

- Test 1: Addition of NaOH(aq) gives a green precipitate that turns brown on standing.
- Test 2: Addition of dilute HNO_3 followed by aqueous $\text{Ba(NO}_3)_2$ gives a dense white precipitate.

(a) Identify the cation present in salt X from Test 1, justifying your deduction. [3]

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(b) Identify the anion present in salt X from Test 2, justifying your deduction. [3]

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(c) Deduce the chemical formula of salt X. [2]

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(d) Write the ionic equation for the reaction occurring in Test 2. [2]

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Qualitative Analysis Notes

Tables for the reactions of aqueous cations, anions, and tests for gases (Cambridge 9701 Syllabus Reference)

1. Reactions of Aqueous Cations

Cation	Reaction with NaOH(aq)	Reaction with NH ₃ (aq)
Aluminium, Al ³⁺	White ppt., soluble in excess giving a colorless solution	White ppt., insoluble in excess
Ammonium, NH ₄ ⁺	Ammonia gas produced on warming (turns damp red litmus blue)	No reaction
Barium, Ba ²⁺	Faint white ppt. or no ppt. (high [Ba ²⁺])	No ppt.
Calcium, Ca ²⁺	White ppt. with high [Ca ²⁺], insoluble in excess	No ppt. with dilute solutions
Chromium(III), Cr ³⁺	Grey-green ppt., soluble in excess giving dark green solution	Grey-green ppt., insoluble in excess
Copper(II), Cu ²⁺	Pale blue ppt., insoluble in excess	Pale blue ppt., soluble in excess giving intense deep blue solution
Iron(II), Fe ²⁺	Green ppt., insoluble in excess; turns brown on standing near surface	Green ppt., insoluble in excess; turns brown on standing
Iron(III), Fe ³⁺	Red-brown ppt., insoluble in excess	Red-brown ppt., insoluble in excess
Magnesium, Mg ²⁺	White ppt., insoluble in excess	White ppt., insoluble in excess
Manganese(II), Mn ²⁺	Off-white ppt., insoluble in excess; rapidly darkens to brown	Off-white ppt., insoluble in excess; darkens to brown
Zinc, Zn ²⁺	White ppt., soluble in excess giving a colorless solution	White ppt., soluble in excess giving a colorless solution

2. Reactions of Aqueous Anions

Ion	Test	Observation
Carbonate, CO ₃ ²⁻	Add dilute acid	Effervescence; gas turns limewater milky (CO ₂)
Chloride, Cl ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	White ppt., soluble in dilute aqueous ammonia
Bromide, Br ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Cream ppt., insoluble in dilute NH ₃ , soluble in conc. NH ₃
Iodide, I ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Yellow ppt., insoluble in dilute and conc. aqueous NH ₃
Nitrate, NO ₃ ⁻ (aq)	Warm with aqueous NaOH, then add Al foil	Ammonia produced; pungent gas turns damp red litmus blue
Sulfate, SO ₄ ²⁻ (aq)	Add dilute HNO ₃ , then aqueous Ba(NO ₃) ₂ / BaCl ₂	Dense white ppt. (insoluble in dilute strong acids)
Sulfite, SO ₃ ²⁻ (aq)	Add dilute acid, warm gently and test gas	Gas produced (SO ₂) turns acidified K ₂ Cr ₂ O ₇ from orange to green

3. Tests for Gases

Gas	Test and Test Result
Ammonia, NH ₃	Turns damp red litmus paper blue; choking pungent smell
Carbon dioxide, CO ₂	Gives a white precipitate with limewater (calcium hydroxide solution turns milky)
Chlorine, Cl ₂	Bleaches damp litmus paper white; pale green choking gas
Hydrogen, H ₂	'Pops' with a lighted wooden splint
Oxygen, O ₂	Relights a glowing wooden splint
Sulfur dioxide, SO ₂	Turns acidified potassium dichromate(VI) paper from orange to green

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical: Systematic Qualitative Analysis — Cations, Anions & Gases

Question 1: Systematic Differentiation of Amphoteric Metal Ions: Al^{3+} , Zn^{2+} , and Mg^{2+} [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Al^{3+} : White ppt soluble in excess NaOH , insoluble in excess NH_3 [1.5]; Zn^{2+} : White ppt soluble in excess of both NaOH and NH_3 [1.5]; Mg^{2+} : White ppt insoluble in excess of both [1].	[4]
(b)	$[\text{Zn}(\text{NH}_3)_4]^{2+}$ (tetraamminezinc(II) ion) [2].	[2]
(c)	$\text{Al}(\text{OH})_3(\text{s}) + \text{OH}^-(\text{aq}) \rightarrow [\text{Al}(\text{OH})_4]^-(\text{aq})$ [2].	[2]
(d)	$\text{Mg}(\text{OH})_2$ is a purely basic hydroxide with strong ionic character; it lacks amphoteric character and cannot accept further OH^- ions [2].	[2]

Question 2: Precipitation and Complex Ion Ligand Substitution of Copper(II) Ions [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Pale blue precipitate forms [1]; precipitate is insoluble in excess NaOH [1].	[2]
(b)	Pale blue precipitate forms with dropwise NH_3 [1]; dissolves in excess NH_3 to give an intense deep royal blue solution [2].	[3]
(c)	Formula: $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ [1]; Geometry: Octahedral (distorted octahedral) [1].	[2]
(d)	White / off-white precipitate of copper(I) iodide (CuI) [1.5]; in a dark brown solution of iodine (I_2) [1.5].	[3]

Question 3: Redox Precipitation and Aerial Oxidation of Iron(II) Hydroxide [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Dirty green / dark green precipitate of $\text{Fe}(\text{OH})_2$ formed [2].	[2]
(b)	The surface of the green precipitate turns reddish-brown [2].	[2]
(c)	$\text{Fe}(\text{OH})_2$ is oxidized by atmospheric dissolved oxygen to iron(III) hydroxide, $\text{Fe}(\text{OH})_3$ (red-brown) [2]; iron is oxidized from +2 to +3 [1].	[3]
(d)	$4\text{Fe}(\text{OH})_2(\text{s}) + \text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 4\text{Fe}(\text{OH})_3(\text{s})$ [3].	[3]

Question 4: Alkaline Oxidation of Chromium(III) Ions by Hydrogen Peroxide [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	$[\text{Cr}(\text{OH})_6]^{3-}$ or $[\text{Cr}(\text{OH})_4]^-$ (tetrahydroxochromate(III)) [2].	[2]
(b)	Effervescence (O_2 gas) and the solution turns bright yellow [2].	[2]
(c)	Chromate(VI) ion, CrO_4^{2-} [2]; Oxidation state of Cr = +6 [1].	[3]
(d)	$2[\text{Cr}(\text{OH})_4]^- (\text{aq}) + 3\text{H}_2\text{O}_2(\text{aq}) + 2\text{OH}^- (\text{aq}) \rightarrow 2\text{CrO}_4^{2-} (\text{aq}) + 8\text{H}_2\text{O}(\text{l})$ [3].	[3]

Question 5: Chemical Discrimination of Sulfate (SO_4^{2-}) and Sulfite (SO_3^{2-}) Ions [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Dense white precipitate forms with BaCl_2 [1.5]; precipitate remains completely insoluble upon adding dilute HCl [1.5].	[3]
(b)	White precipitate forms with BaCl_2 [1]; precipitate dissolves completely in dilute HCl with effervescence [1.5]; choking gas evolved turns acidified potassium dichromate(VI) paper from orange to green [1.5].	[4]
(c)	Sulfur dioxide, SO_2 [1]; $\text{Cr}_2\text{O}_7^{2-} (\text{aq}) + 3\text{SO}_2(\text{g}) + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l})$ [2].	[3]

Question 6: Reduction of Nitrate Ions by Aluminium in Alkaline Solution [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Vigorous effervescence / aluminium dissolves [1]; choking pungent gas turns damp red litmus paper blue (ammonia, NH ₃) [2].	[3]
(b)	Nitrogen is reduced from +5 (in NO ₃ ⁻) to -3 (in NH ₃) [1.5]; Aluminium is oxidized from 0 (in Al) to +3 (in [Al(OH) ₄] ⁻) [1.5].	[3]
(c)	3NO ₃ ⁻ (aq) + 8Al(s) + 5OH ⁻ (aq) + 18H ₂ O(l) → 3NH ₃ (g) + 8[Al(OH) ₄] ⁻ (aq) [2].	[2]
(d)	Water is required to dissolve NH ₃ gas to produce OH ⁻ ions that trigger the alkaline color change of litmus: NH ₃ + H ₂ O ⇌ NH ₄ ⁺ + OH ⁻ [2].	[2]

Question 7: Analytical Verification of Carbonate Ions and Limewater Precipitation [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Rapid effervescence / vigorous bubbling [1]; solid dissolves to form a clear colorless solution [1].	[2]
(b)	Initial: Limewater turns milky / white precipitate of CaCO ₃ forms [2]; Prolonged bubbling: The milky precipitate redissolves to give a clear colorless solution [1] due to the formation of soluble calcium hydrogencarbonate, Ca(HCO ₃) ₂ [1].	[4]
(c)	CaCO ₃ (s) + CO ₂ (g) + H ₂ O(l) → Ca ²⁺ (aq) + 2HCO ₃ ⁻ (aq) [2].	[2]
(d)	Add MgSO ₄ (aq): Na ₂ CO ₃ forms a white precipitate of MgCO ₃ immediately at room temperature [1]; NaHCO ₃ forms no precipitate in the cold, but gives a precipitate upon boiling [1].	[2]

Question 8: Comprehensive Laboratory Protocol for Analytical Gas Detection [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Insert a lighted wooden splint into the gas: 'pops' with a squeaky sound [2].	[2]
(b)	Insert a glowing wooden splint into the gas: splint relights [2].	[2]
(c)	Hold damp blue litmus paper in the gas: turns momentarily red, then is bleached completely white [2].	[2]
(d)	Pass gas over paper soaked in acidified potassium dichromate(VI): color changes from orange to green [2].	[2]
(e)	Hold damp red litmus paper in the gas: turns blue [2].	[2]

Question 9: Systematic Confirmation of the Ammonium Ion (NH₄⁺) in Unknown Salts [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Add aqueous NaOH to the sample in a test-tube [1]; warm gently with a Bunsen flame [1]; test vapors with damp red litmus paper held at tube mouth [1]; damp red litmus turns blue and a choking pungent smell of ammonia is detected [1].	[4]
(b)	NH ₄ ⁺ (aq) + OH ⁻ (aq) → NH ₃ (g) + H ₂ O(l) [2].	[2]
(c)	Splashes of alkaline NaOH liquid on the tube wall would turn the paper blue directly, producing a false positive result [2].	[2]
(d)	Aqueous ammonia already contains NH ₃ molecules and NH ₄ ⁺ ions in dynamic equilibrium, making it impossible to detect NH ₄ ⁺ from the salt [2].	[2]

Question 10: Deductive Identification of an Unknown Binary Inorganic Salt (Salt X) [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Cation: Iron(II), Fe ²⁺ [1]; Green precipitate of Fe(OH) ₂ insoluble in excess NaOH, oxidizing to brown Fe(OH) ₃ in air [2].	[3]
(b)	Anion: Sulfate, SO ₄ ²⁻ [1]; Forms insoluble dense white precipitate of BaSO ₄ with acidified barium nitrate [2].	[3]
(c)	FeSO ₄ (iron(II) sulfate) [2].	[2]
(d)	Ba ²⁺ (aq) + SO ₄ ²⁻ (aq) → BaSO ₄ (s) [2].	[2]